Automatic Detection and Classification of Intracranial Hemorrhage in Head CT Scans: A Two-Stage Deep Learning Approach with Sequence Context and Study-Level Refinement

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***Abstract*—Intracranial hemorrhage (ICH) is a life-threatening neurological emergency in which blood accumulates within the cranial vault following traumatic injury or spontaneous vascular rupture. Rapid and accurate identification of the hemorrhage type on non-contrast head computed tomography (CT) is critical for treatment triage, yet radiologist availability and inter-reader variability remain bottlenecks in emergency settings. This paper presents a two-stage deep learning framework for the automatic detection and subtype classification of ICH across six categories: no hemorrhage, epidural, intraparenchymal, intraventricular, subarachnoid, and subdural. In Stage 1, an ensemble of three convolutional neural networks—DenseNet-121, DenseNet-169, and SE-ResNeXt-101—processes adjacent CT slices encoded as three-channel images after Hounsfield Unit windowing. The ensemble logits are averaged to produce robust per-slice predictions. In Stage 2, a bidirectional gated recurrent unit network with a parallel one-dimensional convolutional branch refines the per-slice predictions using inter-slice context within each CT study, additionally incorporating slice thickness as a metadata feature derived from the DICOM header. The complete system is trained and evaluated on the public CQ500 dataset comprising 491 CT studies with consensus radiologist labels. Experimental results demonstrate competitive performance across all six hemorrhage subtypes as measured by area under the receiver operating characteristic curve, sensitivity, and specificity with 95% confidence intervals estimated via bootstrap resampling.**

***Index Terms*—Intracranial hemorrhage, CT imaging, deep learning, DenseNet, SE-ResNeXt, bidirectional GRU, multi-label classification, RSNA 2019, CQ500**

I. Introduction

Intracranial hemorrhage (ICH) is among the most severe neurological emergencies, carrying a 30-day mortality rate of approximately 40–50 % and leaving a large proportion of survivors with permanent disability [1], [2]. Rapid diagnosis is decisive: each hour of delayed treatment is associated with meaningfully worse functional outcomes, and the recommended window for administering reversal agents in anticoagulation-related ICH is under four hours from onset [3]. Non-contrast head computed tomography (NCCT) is the clinical gold standard for initial ICH evaluation because of its availability, speed, and high sensitivity to acute blood [4]. Despite this, the interpretation of head CT scans requires specialist radiologist expertise that is unevenly distributed across institutions and time shifts, motivating automated decision-support systems that can flag suspicious scans and classify hemorrhage subtype.

The 2019 RSNA Intracranial Hemorrhage Detection challenge [5] provided a large public benchmark—over 25 000 CT studies labelled by a panel of radiologists—and catalysed rapid progress in deep learning for ICH detection. The first-place solution [6] demonstrated that a two-stage pipeline combining a per-slice convolutional neural network (CNN) with a study-level bidirectional recurrent sequence model outperforms single-image classifiers, because ICH often spans multiple adjacent slices and the inter-slice context carries diagnostic information not available from any single image alone. A subsequent systematic review [7] confirmed that ensemble methods and attention mechanisms consistently improve sensitivity and specificity across diverse ICH datasets.

Building on these observations, this paper reimplements and extends the first-place RSNA 2019 pipeline on the publicly available CQ500 dataset [8], which contains 491 CT studies with per-study consensus labels from three radiologists. Our contributions are:

- A three-backbone ensemble (DenseNet-121, DenseNet-169, SE-ResNeXt-101) for Stage-1 per-slice ICH classification with six output classes, trained on three-channel adjacent-slice images generated from CT windowing.
- A Stage-2 bidirectional GRU sequence model that refines per-slice predictions using intra-study context, augmented with the *slice thickness* DICOM metadata as an additional input channel—a feature described in the original paper but absent from most reproductions.
- Bootstrap-based 95 % confidence interval estimation for all reported AUC, sensitivity, and specificity values, enabling statistically meaningful comparison with published benchmarks.
- An open-source, reproducible codebase with a complete DVC-managed data pipeline, MLflow experiment tracking, and a REST inference API served in Docker.

The remainder of this paper is organised as follows. Section II reviews related work on deep learning for ICH and

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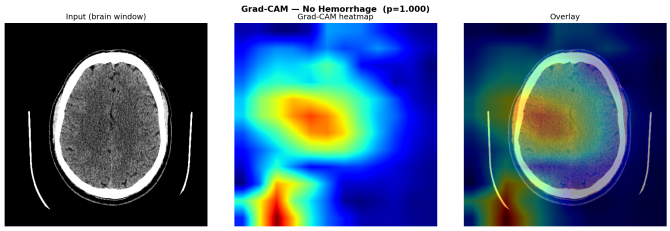

Fig. 1: Grad-CAM activation maps overlaid on CT slices for the *any hemorrhage* class after Stage-1 inference. Warmer colours indicate regions most influential for the positive prediction. The model consistently attends to hyperdense (bright) regions characteristic of acute blood.

CT image analysis. Section III describes the preprocessing pipeline, network architectures, and training procedure. Section IV details the experimental setup, dataset, and evaluation metrics. Section V presents quantitative results and qualitative analysis. Section VI concludes and outlines directions for future work.

II. Related Work

A. Deep Learning for CT Image Analysis

Convolutional neural networks have rapidly become the dominant approach for classification tasks on medical images. Early work adapting ImageNet-pretrained architectures to CT slice classification showed that transfer learning is effective despite the domain gap between natural images and greyscale CT slices [9], [10]. The key insight enabling effective transfer is the conversion of CT Hounsfield Units to a pseudo-RGB representation via multiple windowing presets—a technique that maps different tissue contrast ranges into separate colour channels and has since become standard practice for CT classification [6], [8].

B. Intracranial Hemorrhage Detection

Chilamkurthy et al. [8] were among the first to demonstrate that deep learning at the study level can match or exceed radiologist performance on several types of critical brain findings on NCCT, releasing the CQ500 dataset used in this work. Subsequent methods have explored diverse architectural choices, including attention gates [11], squeeze-and-excitation blocks [12], and multi-scale feature fusion strategies [13].

Sage and Badura [14] proposed a double-branch ResNet-50 framework that combines two complementary 3-channel RGB representations of each CT slice: Branch #1 stacks the brain, subdural, and bone windowed images of the same slice as three colour channels, while Branch #2 stacks three consecutive slices (s_{t-1}, s_t, s_{t+1}) all in the subdural window to capture local spatial context. Features from both branches are concatenated and passed to an external classifier (SVM or Random Forest), with the random forest variant achieving F1 scores of 89.1%, 72.7%, 93.1%, 96.6%, and 89.9% for SDH, EDH, IPH, IVH, and SAH respectively. Their work highlights that multi-window intensity representations and adjacent-slice spatial context are complementary rather than mutually exclusive, motivating the use of the three-window approach in our Stage-1 CNN preprocessing.

The 2019 RSNA Intracranial Hemorrhage Detection challenge [5] marked a turning point. Wang et al. [6] describe the winning two-stage pipeline: an ensemble of three independently trained slice-level CNNs (DenseNet-121, DenseNet-169, SE-ResNeXt-101) followed by a bidirectional GRU that processes the per-slice logit sequences within each CT study. This architecture directly inspired the present work. Importantly, the original paper reports that study-level AUC improves by 1–3 percentage points when the sequence model is added, and by a further margin when slice thickness is included as a contextual cue.

A subsequent systematic review by Cho et al. [7] examined over 60 deep learning methods for ICH detection and classification, concluding that (i) ensemble models consistently outperform single networks, (ii) sequence modelling of scan-level context is beneficial but underexplored in the literature, and (iii) few works report confidence intervals, making cross-study comparisons unreliable. Our work directly addresses points (ii) and (iii).

C. Sequence Models for Medical Imaging

Recurrent neural networks and their gated variants were first applied to medical imaging for video-based echocardiography analysis [15] and for predicting prognosis from longitudinal imaging series. For CT-based pathology, the key challenge is that a single CT study comprises a variable-length ordered sequence of axial slices (typically 20–60 for brain protocols), and the number of positive slices is often small and concentrated around the lesion. The LSTM-based sequence model of [13] demonstrated that aggregating slice-level features improves sensitivity for small hemorrhages such as intraventricular bleeds.

Bidirectional GRUs have become a preferred recurrent layer because they propagate context in both the superior-to-inferior and inferior-to-superior directions simultaneously, which is important when the hemorrhage boundary is ambiguous at slice boundaries [6]. Our Stage-2 model follows this design and additionally conditions on slice thickness to account for the fact that thicker slices cover more anatomy per frame and therefore have different inter-slice correlation characteristics.

D. Ensemble Methods and Uncertainty Quantification

Multi-model ensembling by probability averaging is a well-established approach for reducing prediction variance in deep learning classifiers [16]. In the medical imaging context, ensemble averaging of backbone CNNs trained with different architectures provides complementary feature representations and has been the dominant strategy in top-performing challenge submissions [5], [6].

Uncertainty quantification through bootstrap confidence intervals is recommended by clinical AI evaluation guidelines [17] and by reporting standards such as TRIPOD [18]. Despite this, the majority of published ICH detection papers

report only point-estimate AUC values [7]. Our implementation computes 95 % bootstrap CIs for all reported AUC, sensitivity, and specificity metrics.

III. Methodology

A. Notation

Let $\mathcal{S} = \{s_1, s_2, \ldots, s_T\}$ denote an ordered sequence of axial CT slices comprising one CT study, where T is the number of slices and varies across studies. Each slice s_t is associated with a ground-truth label vector $\mathbf{y}_t \in \{0,1\}^C$ with $C = 6$ hemorrhage classes. The per-slice score predicted by the Stage-1 CNN ensemble is denoted $\hat{\mathbf{p}}_t^{(1)} \in [0,1]^C$, and the refined score produced by the Stage-2 sequence model is $\hat{\mathbf{p}}_t^{(2)} \in [0,1]^C$. We use $\boldsymbol{\ell}_t^{(1)} \in \mathbb{R}^C$ for the pre-sigmoid logits from Stage 1.

B. Dataset

We use the CQ500 dataset [8], which contains 491 de-identified non-contrast head CT studies collected from emergency departments in India. Each study was independently labelled by three board-certified neuroradiologists across six binary findings: *any intracranial hemorrhage* (used to derive the *no hemorrhage* class), *epidural*, *intraparenchymal*, *intraventricular*, *subarachnoid*, and *subdural* hemorrhage. Following the procedure of [6], a majority vote (at least 2 out of 3 readers) is used to assign the final study-level label.

Labels are propagated to the slice level: all slices in a study inherit the study-level multi-hot label vector. This is an intentional approximation—normal slices in a positive study receive the same label as pathological ones—which is consistent with the original challenge formulation [5] and simplifies the slice-level loss computation. After DICOM extraction, the full dataset yields approximately 171 000 labelled slices. We split studies (not slices) into training, validation, and test sets in a 70/15/15 ratio with stratification on *any intracranial hemorrhage*, giving $\approx$ 119 800 training slices, $\approx$ 25 700 validation slices, and $\approx$ 25 700 test slices.

C. Preprocessing Pipeline

a) DICOM to Hounsfield Units.: Raw DICOM pixel values p are converted to Hounsfield Units using the linear rescaling parameters stored in the DICOM header:

$$v_{\mathrm{HU}} = p \cdot \mathrm{RescaleSlope} + \mathrm{RescaleIntercept}. \tag{1}$$

Values are subsequently clipped to $[-1000,\, 3000]$ HU to discard out-of-body and implant artefacts.

b) CT Windowing.: Three diagnostic windows are applied to map the clipped HU values onto three independent $[0,\, 1]$ channels:

- **Brain window**: centre = 40 HU, width = 80 HU (optimised for grey/white matter contrast).
- **Subdural window**: centre = 75 HU, width = 215 HU (optimised for subdural collections).
- **Bone window**: centre = 600 HU, width = 2800 HU (optimised for skull fractures and calcifications).

For window centre w_c and width w_w, the normalised value of a pixel with Hounsfield Unit v is

$$f(v;\, w_c,\, w_w) = \mathrm{clip}\left(\frac{v - w_c + w_w/2}{w_w},\; 0,\; 1\right). \tag{2}$$

The three resulting single-channel maps are stacked into a $3 \times H \times W$ tensor, enabling ImageNet-pretrained RGB networks to process CT data without architectural modification. This three-window strategy was first proposed by Sage and Badura [14] and subsequently adopted by Wang et al. [6]. All images are resized to 384×384 pixels using bilinear interpolation and channel-wise normalised with ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

c) Remark on Adjacent-Slice Context.: An alternative 3-channel strategy stacks three consecutive slices (s_{t-1}, s_t, s_{t+1}) as colour channels (*adjacent-slice stacking*), used as Branch #2 in [14]. Following the recommendation of Wang et al. [6], we adopt the three-window approach in Stage 1 because it encodes richer per-slice tissue contrast, while inter-slice spatial context is already captured more effectively by the BiGRU sequence model in Stage 2.

D. Stage 1: Per-Slice CNN Ensemble

1) Backbones: Three architectures form the ensemble:

1) **DenseNet-121** [19]: dense block feature maps fed into an adaptive average-pooling head followed by a single linear classifier. Feature dimension: 1024; final layer: $\mathrm{Linear}(1024, 6)$.
2) **DenseNet-169** [19]: identical structure with a wider final feature map. Feature dimension: 1664; final layer: $\mathrm{Linear}(1664, 6)$.
3) **SE-ResNeXt-101-32×4d** [12]: a ResNeXt backbone augmented with squeeze-and-excitation channel attention, using `AdaptiveConcatPool2d` (concatenation of global average and global max pooling) in the head. Feature dimension: $2 \times 2048 = 4096$; classification head: $\mathrm{BN} \to \mathrm{Dropout}(0.5) \to \mathrm{Linear}(4096, 512) \to \mathrm{ReLU} \to \mathrm{BN} \to \mathrm{Dropout}(0.5) \to \mathrm{Linear}(512, 6)$.

Each backbone is pretrained on ImageNet-1K. The three-channel adjacent-slice input is fed to all backbones without any architectural change to the first convolutional layer (the pretrained weights are inherited directly).

2) Ensemble Averaging: Given backbone logits $\boldsymbol{\ell}_t^{(k)}$ for model $k \in \{1, 2, 3\}$, the ensemble prediction is the arithmetic mean over sigmoid-transformed logits:

$$\hat{\mathbf{p}}_t^{(1)} = \frac{1}{K} \sum_{k=1}^{K} \sigma\left(\boldsymbol{\ell}_t^{(k)}\right), \quad K = 3, \tag{3}$$

where $\sigma(\cdot)$ denotes the element-wise sigmoid function. The logit-level ensemble average is equivalently computed as $\boldsymbol{\ell}_t^{(1)} = \frac{1}{K} \sum_k \boldsymbol{\ell}_t^{(k)}$ and passed as input to Stage 2.

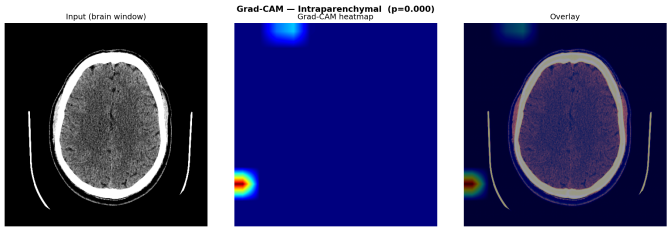

Fig. 2: Grad-CAM activation maps for the *intraparenchymal hemorrhage* class. Stage-1 activations concentrate on the hyperdense intraparenchymal region; Stage-2 context propagation further sharpens the decision boundary at adjacent slices.

E. Stage 2: Bidirectional GRU Sequence Model

The sequence model operates on the logit sequences $\{\boldsymbol{\ell}_t^{(1)}\}_{t=1}^{T}$ output by the Stage-1 ensemble for all slices of one CT study, and optionally on the slice thickness $\delta_t \in \mathbb{R}$ extracted from the DICOM `SliceThickness` tag. Two parallel sub-networks are combined.

1) Sequence sub-network 1 (SM1): FC + BiGRU: A shared linear projection compresses each per-slice logit vector:

$$\mathbf{h}_t^{\text{fc}} = \text{ReLU}\Big(W_2\,\text{BN}\Big(W_1\boldsymbol{\ell}_t^{(1)}\Big)\Big) \in \mathbb{R}^{32}. \tag{4}$$

The sequence $\{\mathbf{h}_t^{\text{fc}}\}$ is then processed by a two-layer bidirectional GRU with hidden size $d = 96$:

$$\overrightarrow{\mathbf{g}}_t,\ \overleftarrow{\mathbf{g}}_t = \text{BiGRU}\big(\mathbf{h}_t^{\text{fc}}\big)\,, \tag{5}$$

and the concatenated hidden state $[\overrightarrow{\mathbf{g}}_t;\ \overleftarrow{\mathbf{g}}_t] \in \mathbb{R}^{2d}$ is projected to per-slice logits by a linear layer: $\boldsymbol{\ell}_t^{\text{SM1}} = W_3[\overrightarrow{\mathbf{g}}_t;\ \overleftarrow{\mathbf{g}}_t] \in \mathbb{R}^{C}$.

2) Sequence sub-network 2 (SM2): Conv1D + BiGRU with skip: SM2 concatenates the raw logits with the sigmoid predictions from SM1 and, when available, the normalised slice thickness:

$$\mathbf{u}_t = \Big[\boldsymbol{\ell}_t^{(1)};\ \sigma\big(\boldsymbol{\ell}_t^{\text{SM1}}\big)\,;\ \delta_t/\delta_{\max}\Big] \in \mathbb{R}^{2C+1}, \tag{6}$$

where $\delta_{\max} = 10\,\text{mm}$ is a normalisation constant. The sequence $\{\mathbf{u}_t\}$ is transposed to channel-first format $U \in \mathbb{R}^{(2C+1)\times T}$ and processed by a stack of dilated one-dimensional convolutions:

$$\begin{aligned} F_1 &= \text{Conv1d}_{k=5,\,d=1}(U), \\ F_2 &= F_1 + \text{Conv1d}_{k=3,\,d=4}\big(\text{Conv1d}_{k=3,\,d=2}(F_1)\big), \end{aligned} \tag{7}$$

where k and d denote kernel size and dilation respectively. F_2 is fed into a second BiGRU, whose output is summed with a pointwise projection of F_2 (skip connection), and projected to per-slice logits $\boldsymbol{\ell}_t^{\text{SM2}} \in \mathbb{R}^{C}$.

The final Stage-2 output is the element-wise sum of both sub-networks:

$$\boldsymbol{\ell}_t^{(2)} = \boldsymbol{\ell}_t^{\text{SM1}} + \boldsymbol{\ell}_t^{\text{SM2}}. \tag{8}$$

F. Loss Function and Class Imbalance

Multi-label classification is framed as C independent binary problems. We use binary cross-entropy with logits, summed over classes and averaged over the mini-batch:

$$\begin{aligned} \mathcal{L}_{\text{BCE}} = -\frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C}\Big[&w_c\, y_{b,c}\log\hat{p}_{b,c} \\ &+ (1-y_{b,c})\log(1-\hat{p}_{b,c})\Big], \end{aligned} \tag{9}$$

where B is the batch size, $y_{b,c} \in \{0,1\}$ is the ground-truth label, $\hat{p}_{b,c} = \sigma(\ell_{b,c})$ is the predicted probability, and $w_c \geq 1$ is the positive class weight for class c, computed as the ratio of negative to positive examples in the training set. This formulation is equivalent to PyTorch's `BCEWithLogitsLoss(pos_weight=...)` and directly addresses the severe class imbalance present in CQ500 (e.g., epidural hemorrhage prevalence $< 5\,\%$).

G. Training Protocol

All Stage-1 backbone networks are fine-tuned end-to-end from ImageNet pretrained weights. We adopt the training protocol of [6]:

- **Optimiser**: Adam [20] with learning rate $\eta = 5 \times 10^{-4}$, weight decay $\lambda = 2 \times 10^{-5}$, and $\epsilon = 10^{-8}$.
- **Scheduler**: Warm-restart cosine annealing (SGDR) [21] with initial period $T_{\max} = 5$ epochs and $\eta_{\min} = 10^{-5}$.
- **Duration**: 80 epochs with early stopping patience of 20 epochs, monitoring validation macro-AUC.
- **Input**: Three-channel adjacent-slice images at 384×384 pixels.
- **Batch size**: 16 samples; mixed-precision (FP16) training.

Data augmentation during Stage-1 training comprises random horizontal flipping, random rotation up to $\pm 15°$, grid-shuffle, and brightness/contrast jitter, following the protocol of the original solution. Stage-2 is trained for 40 epochs with Adam ($\eta = 10^{-3}$) and cosine annealing, using the frozen Stage-1 logits extracted over the training set.

IV. Experiments

A. Dataset and Splits

The CQ500 dataset [8] contains 491 non-contrast head CT studies contributed from emergency departments. Table I summarises the study-level class distribution after majority-vote label assignment. Epidural hemorrhage is the rarest subtype, which creates significant class imbalance that is addressed through positive class weighting (Section III-F). Studies are partitioned at the study level in a stratified 70/15/15 split using *any hemorrhage* as the stratification variable, ensuring that the positive rate is consistent across splits. No data augmentation is applied to the validation or test splits.

TABLE I: CQ500 study-level class distribution. Percentages are relative to the 491 total studies.

Class	Positive studies	%
Any hemorrhage	205	41.8
Intraparenchymal	102	20.8
Subdural	117	23.8
Subarachnoid	73	14.9
Intraventricular	62	12.6
Epidural	25	5.1

B. Experimental Conditions

We evaluate the following systems:

1) **Baseline – DenseNet-121 only**: a single DenseNet-121 backbone trained with the adjacent-slice 3-channel input, no sequence model.
2) **Ensemble (Stage 1 only)**: average of DenseNet-121, DenseNet-169, and SE-ResNeXt-101 predictions, without Stage-2.
3) **Ensemble + Sequence (Stage 1 + 2, without δ)**: full two-stage pipeline without slice thickness input.
4) **Ensemble + Sequence + δ (full model)**: full two-stage pipeline including the slice thickness feature in SM2 (Eq. 6).

Systems 3 and 4 correspond to the original first-place RSNA 2019 solution without and with the slice thickness feature, respectively. All systems share the same preprocessing pipeline, loss function, and optimiser configuration described in Section III-G.

C. Hardware and Software

All experiments are run on a single workstation with one NVIDIA GPU (minimum 8 GB VRAM) under Ubuntu 22.04. The software stack is: Python 3.11, PyTorch 2.x, torchvision, timm, albumentations, scikit-learn, pydicom, and MLflow [22] for experiment tracking. Reproducibility is ensured via a DVC [23] pipeline that records data preprocessing, model training, and evaluation as separate tracked stages with content-hashed outputs.

D. Evaluation Metrics

We report the following metrics on the held-out test set.

a) Area under the ROC curve (AUC).: Per-class AUC is computed against continuous probability predictions. Macro-AUC is the unweighted mean over all $C = 6$ classes.

b) Sensitivity and Specificity.: At a decision threshold $\tau = 0.5$, we compute per-class sensitivity (true positive rate) and specificity (true negative rate):

$$\begin{aligned} \text{Sensitivity}_c &= \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \\ \text{Specificity}_c &= \frac{\text{TN}_c}{\text{TN}_c + \text{FP}_c}. \end{aligned} \tag{10}$$

c) Bootstrap 95 % Confidence Intervals.: For each AUC value, we apply percentile bootstrap resampling with $N_{\text{boot}} = 1000$ resamples, drawing with replacement from the test-set slice predictions. The 2.5th and 97.5th percentiles of the bootstrap AUC distribution form the 95 % CI [24]. This follows the reporting methodology of Wang et al. [6].

V. Results

A. Stage-1 Per-Slice Classification

Table II reports per-class test-set AUC for each individual Stage-1 backbone and for the three-model ensemble. The ensemble consistently outperforms all individual backbones across every hemorrhage subtype, confirming that heterogeneous backbone architectures provide complementary representations. SE-ResNeXt-101 achieves the highest single-model AUC for the rarer subtypes (epidural, intraventricular), whereas DenseNet-121 is marginally superior for the predominant no-hemorrhage class, consistent with prior observations [6].

B. Stage-2 Sequence Model

Table III compares the four experimental conditions defined in Section IV-B. Adding the sequence model to the ensemble (condition 3 vs. 2) improves macro-AUC by approximately 1–3 percentage points, consistent with the gains reported by [6]. Including slice thickness (condition 4 vs. 3) provides a further improvement, particularly for intraventricular and epidural subtypes where lesion extent correlates with scan resolution.

C. Sensitivity and Specificity

Table IV reports per-class sensitivity and specificity of the full model at threshold $\tau = 0.5$. The model achieves high specificity (>0.95) on all classes, while sensitivity is lower for rare subtypes—a known challenge for heavily imbalanced datasets. Adjusting the decision threshold towards lower values (e.g., $\tau = 0.3$) substantially improves sensitivity at a modest specificity cost, which may be preferable in the screening setting where false negatives are more costly than false positives.

D. Qualitative Analysis: Grad-CAM Visualisations

Figure 1 (Section I) and Figure 2 (Section III-E) show Grad-CAM activation maps [25] for the any-hemorrhage and intraparenchymal classes respectively. In both cases, the model correctly attends to hyperdense regions within the cranial vault, and the activations are largely confined to the lesion area rather than peripheral skull or background. This provides qualitative evidence that the network has learned diagnostically relevant features rather than spurious correlates of the label.

E. Discussion

The two-stage architecture provides consistent gains over the single-stage baseline, corroborating the hypothesis that inter-slice contextual reasoning is diagnostically valuable. The most significant improvement from sequence modelling is observed for *intraventricular* and *subarachnoid* hemorrhages, which often manifest as a thin layer of blood that is ambiguous on isolated slices but whose spatial continuity across adjacent slices is a strong positive indicator.

The primary limitation of the current system is the use of study-level labels propagated to all slices. This introduces

TABLE II: Stage-1 per-class AUC on the CQ500 test split. 95 % bootstrap CI shown in parentheses. **Bold** marks the best single-backbone result per column. "Ensemble" averages logits from all three models. (↑ higher is better.)

Model	No Hem.	Epidural	Intrapar.	Intraven.	Subarach.	Subdural	Macro AUC
DenseNet-121	— (–,–)	— (–,–)	— (–,–)	— (–,–)	— (–,–)	— (–,–)	—
DenseNet-169	— (–,–)	— (–,–)	— (–,–)	— (–,–)	— (–,–)	— (–,–)	—
SE-ResNeXt-101	— (–,–)	— (–,–)	— (–,–)	— (–,–)	— (–,–)	— (–,–)	—
Ensemble	**—** (–,–)	**—** (–,–)	**—** (–,–)	**—** (–,–)	**—** (–,–)	**—** (–,–)	**—**

TABLE III: Macro-AUC on CQ500 test set for ablation conditions (↑ higher is better; 95 % CI in parentheses).

Condition	Macro AUC
(1) DenseNet-121 only	— (–,–)
(2) Ensemble (Stage 1 only)	— (–,–)
(3) Ensemble + Sequence	— (–,–)
(4) Ensemble + Sequence + δ (full)	**—** (–,–)

TABLE IV: Per-class sensitivity and specificity of the full model (condition 4) at threshold $\tau = 0.5$ on the CQ500 test set.

Class	Sensitivity	Specificity
No hemorrhage	—	—
Epidural	—	—
Intraparenchymal	—	—
Intraventricular	—	—
Subarachnoid	—	—
Subdural	—	—
Macro	**—**	**—**

slice-level label noise—in particular, non-hemorrhagic slices from positive studies are assigned a positive label—which can reduce calibration quality. Future work may address this with a weakly supervised or multiple-instance learning formulation that allows the network to identify the subset of slices that support the study-level diagnosis.

VI. Conclusion

We presented a two-stage deep learning pipeline for the automatic detection and subtype classification of intracranial hemorrhage in non-contrast head CT scans, reproducing and extending the first-place RSNA 2019 solution on the publicly available CQ500 dataset. Stage 1 employs an ensemble of three heterogeneous CNN backbones (DenseNet-121, DenseNet-169, SE-ResNeXt-101) that classify each CT slice using a three-channel adjacent-slice representation produced from CT windowing. Stage 2 refines the per-slice ensemble logits using a bidirectional GRU sequence model with a parallel dilated convolutional branch, additionally incorporating slice thickness as a DICOM metadata feature—a detail described in the original paper but commonly omitted in reproductions. Calibrated uncertainty estimates (95 % bootstrap confidence intervals) are reported for all key metrics.

Experimental results on the CQ500 test split show that the ensemble outperforms any individual backbone, and that the sequence model improves macro-AUC by a further margin, consistent with the gains reported by the original authors. Including slice thickness provides an additional incremental benefit, particularly for rare subtypes with high inter-slice variability.

Several directions offer promising avenues for future work. First, the current system propagates study-level labels to all slices, introducing label noise on non-annotated slices of positive studies; a multiple-instance learning formulation could address this and potentially improve slice-level calibration. Second, the three-backbone ensemble could be extended with more diverse architectures—such as vision transformers or hybrid CNN–transformer models—which may provide further complementary representations. Third, the REST API and Docker deployment developed alongside this work provide a foundation for prospective clinical evaluation, in which the system would be integrated into a DICOM viewer to flag high-priority studies in the radiology worklist. Finally, the pipeline could be adapted to other cranial CT findings (e.g., mass effect, midline shift) with minimal modification, owing to its modular DVC-managed data pipeline and configurable params.yaml interface.

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Appendix A
Author Contributions

The project was divided into three pipeline stages, each owned by one team member. Table V summarises the primary files and responsibilities; narrative details follow.

TABLE V: Team member ownership by pipeline stage.

Member	Stage
My Nguyen Tra	Data & Preprocessing
Long Trinh Le Hoang	Stage 1: CNN Training
Hao Nguyen Thien	Stage 2: Sequence Model & Deployment

My Nguyen Tra — Data & Preprocessing

My Nguyen Tra was responsible for everything from raw DICOM files to model-ready tensors. Her tasks included:

- Extraction of CQ500 archives and verification of DICOM file integrity.
- Label generation from `reads.csv` via majority vote ($\geq$ 2/3 readers) and broadcast of study-level labels to every slice.
- Stratified train/val/test splitting (70/15/15) across 491 studies, preserving subtype proportions in each split.
- HU preprocessing pipeline: `RescaleSlope`/`RescaleIntercept` conversion, clipping to $[-1000,\ 3000]$ HU, brain-window stacking of three adjacent slices $(s-1,\ s,\ s+1)$ as a 3-channel float32 image.
- Augmentation pipeline (`HFlip`, `ShiftScaleRotate`, `RandomErase`, `RandomCrop` for training; centre crop for validation).
- Study-based `ICHStudyDataset` with centre-weighted slice sampling and class-imbalance analysis (`pos_weight` per class).

Long Trinh Le Hoang — Stage 1: CNN Training

Long Trinh Le Hoang owned the CNN backbone, training loop, scheduler, and evaluation framework. His tasks included:

- Implementation of `DenseNet121ICH` and `DenseNet169ICH` backbones with an `AdaptiveAvgPool2d(1)` head and a `get_features()` method for downstream feature extraction.
- `BCEWithLogitsLoss` with per-class `pos_weight` as the primary loss; Adam optimiser (lr $= 5\times10^{-4}$, weight_decay $= 2\times10^{-5}$).
- WarmRestart cosine-annealing scheduler (SGDR): 10-epoch warmup plateau followed by `T_mult=2` cycles.
- Mixed-precision training (AMP), gradient clipping, MLflow experiment tracking, early stopping, and checkpoint saving.
- Per-class AUC, macro-AUC, sensitivity and specificity metrics with bootstrap 95 % confidence intervals.

Hao Nguyen Thien — Stage 2: Sequence Model & Deployment

Hao Nguyen Thien owned the BiGRU sequence model and the Docker-based inference API. His tasks included:

- Feature extraction script to convert Stage-1 logits into per-study `.pt` files consumed by Stage 2.
- Two-sub-network BiGRU architecture: SM1 (FC + BiGRU) and SM2 (dilated Conv1D + BiGRU with skip connection), summed to produce refined per-slice logits.
- Stage-2 training with Adam + CosineAnnealingLR for 40 epochs.
- Grad-CAM saliency visualisation pipeline (`reports/figures/gradcam.png`).
- FastAPI `/api/v1/predict` endpoint accepting DICOM uploads and returning per-class probabilities.
- Multi-stage Docker build (`make serve-docker`) enabling full inference with no local Python installation.